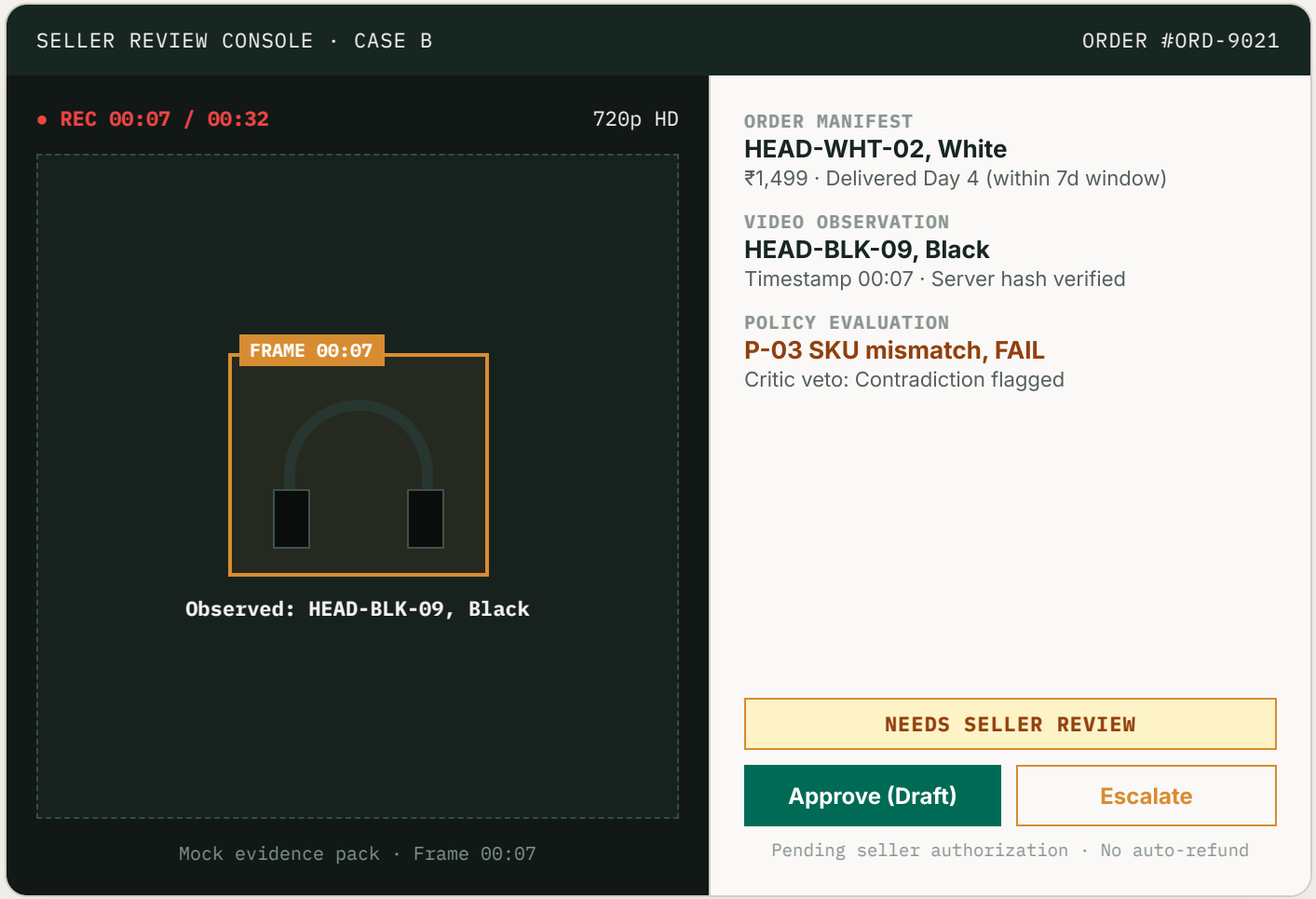
ReturnLens

An evidence-checked returns agent for small direct-to-consumer sellers.

USER Solo D2C merchant managing orders and returns alone.

SCOPE IN 24H Return triage only, mock store data + 1 order-status route.

WHAT IT IS NOT Not another chatbot; not an autonomous refund gate.



Team Rookie · 3 members · PS-04 submission
Deterministic Policy Engine · Grounded Video Proof · 24h Build

3	P-01 to P-08	4	1-Click
MEMBERS	POLICY ENGINE	DEMO CASES	SELLER APPROVAL

PROBLEM STATEMENT · SOLO SELLER BOTTLENECK

The return decision is made from memory, before the evidence is checked.

Today the seller opens many tabs, then decides alone.

01 BUYER CHAT

"Received damaged item"

Unverified text in support inbox.

02 ORDER RECORD

HEAD-WHT-02 · ₹1,499

Database manifest & SKU truth.

⚠ CRITICAL INFORMATION GAP

NO SHARED EVIDENCE RECORD

"Approve, deny, or ask for more evidence?"

03 DELIVERY TRACKING

Delivered 4 days ago

Carrier tracking portal status.

04 RETURN POLICY

7-day window, intact tag

Store PDF terms & conditions.

A wrong decision costs both ways

A wrong refund loses money and stock, an unfair denial loses a repeat customer.

The gap is not a missing portal

It is that no single record ties order, video frame, policy clause, and decision together.

Fraud estimates differ by definition:
9%
(NRF 2025, fraud only) vs
15.14%
(Appriss/Deloitte 2024, fraud + abuse).

CORE DEMONSTRATION · AUDIT PROOF

One frame changes the verdict.

ReturnLens surfaces the contradiction instead of auto-denying.

ORDER TRUTH — **SHOPIFY MANIFEST**

HEAD-WHT-02

Matte White

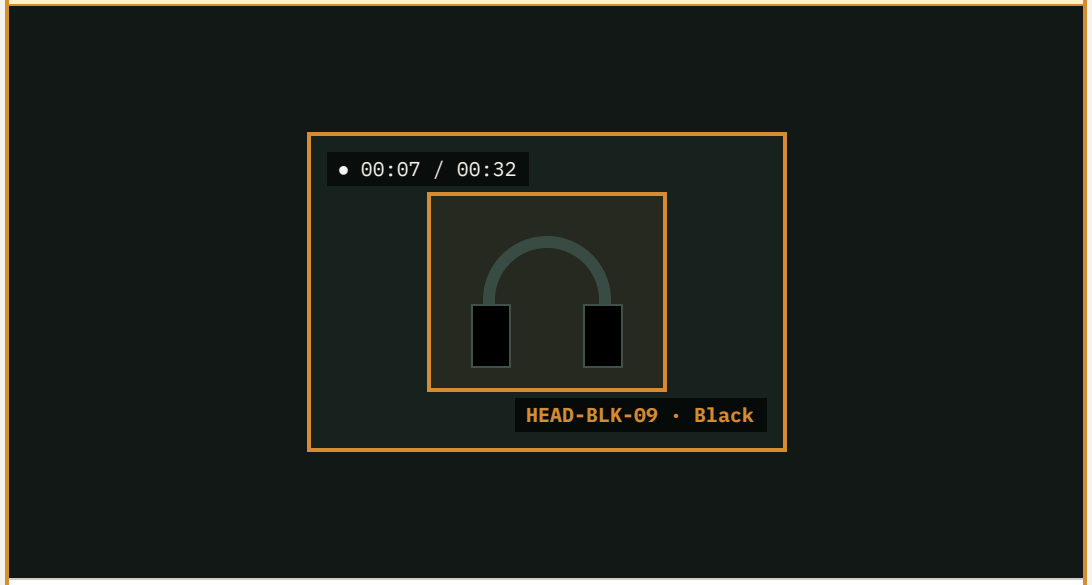
₹1,499 · Delivered 4 days ago

Original retail tag present

Order record specifies White Wireless Over-Ear headset.

VS →

CUSTOMER VIDEO · FRAME 00:07 — **UNBOXING TAKE**



Observed: HEAD-BLK-09, Midnight Black

Mock evidence pack

P-03 SKU mismatch, FAIL → NEEDS SELLER REVIEW

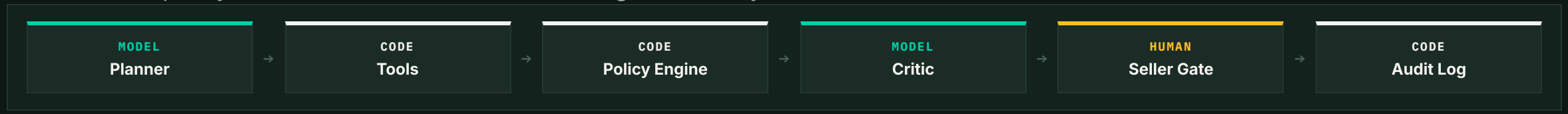
Seller options: Accept as mis-shipment / Ask for outside-box label / Hold for inspection

No label, no refund draft, no denial.

SYSTEM ARCHITECTURE · DETERMINISTIC GOVERNANCE

Policy checks and seller approval control actions

Deterministic policy in code and human authorization govern every state transition.



TOOLS Mock store & shipping APIs + FFmpeg keyframe indexer.

POLICY CLAUSES Deterministic code rules P-01 to P-08, never prompts.

MODEL CALLS Exactly 3 base model calls per case: planner, vision, critic, plus retries.

FRAMES Server-generated with frame_id, timestamp, video hash, and frame hash. The model may only cite frames it was given.

CRITIC The critic reviews cited frames and flags unsupported observations.

FAIL-CLOSED Weak evidence escalates; never auto-deny, never auto-refund.

INJECTION LIMIT Injection cannot change the law, exceed the server cap, or execute an action alone, but it can change an observation field, so the seller must check the frames in the evidence pack.

CONTRADICTION	HARD FAIL	UNCLEAR ONLY	ALL PASS
→ Escalate	→ Escalate	→ Ask More	→ Approve for Review

DEMO SCENARIOS · GROUNDED VERIFICATION

Every decision can be clicked back to a frame, a clause, and a log entry.

Four reproducible benchmark cases prove deterministic safety.

CASE B · AUDIT PROOF (THE CORE BENCHMARK)	
ORDER	**HEAD-WHT-02**, White, ₹1,499, delivered, tag present
FRAME	**Frame 00:07**: **HEAD-BLK-09**, Black
POLICY	**P-03 SKU mismatch**, **FAIL**
RESULT	**ESCALATE** (no label, no refund draft, no denial)
SELLER OPTIONS	Accept as mis-shipment / Ask for outside-box label / Hold for inspection

CASE A · ORD-1042 — READY FOR REVIEW

ORD-1042, ₹1,499, SKU matches, tag present, within window.

Seller approval creates a mock return label and a refund draft in **pending_return_receipt**, not an executed refund.

CASE C · POOR LIGHTING — ASK FOR MORE EVIDENCE

4-second dark video, brightness 28, only 2 usable frames.

Case stays open with an exact re-film checklist.

CASE D · UNBROKEN TAKE — WHY VIDEO BEATS PHOTOS

One continuous take from opening a sealed parcel to showing the label in one continuous sequence.

Prevents photo-swap fraud before mock return label generation.

Staged sample videos, mock data, no real customer data. If the model fails or the quota runs out, escalate. Do not approve without evidence.

EXECUTION PLAN · HACKATHON FEASIBILITY

Buildable in 24 hours by 3 people, if the scope stays narrow.

Project logic is written inside the 24 hours. Before that, only research, design, accounts, and library choice.

0–6h
Skeleton
Mock store & shipping APIs + agent skeleton.

6–12h
Pipeline & Policy
FFmpeg video indexer + P-01 to P-08 code engine.

12–18h
Integration
4 demo cases + independent critic verification.

18–24h
Demo & Polish
Seller console view, fallback tests, video & README.

Person A — BACKEND
Backend, agent graph, policy, critic, audit log.

Person B — VIDEO & UI
Video pipeline, video indexer, seller console, evidence pack view.

Person C — INTEGRATION
Integration, mock data, 4 demo cases, fallback testing, demo video, README.

CUT FIRST
WhatsApp, exchanges, multi-seller, audio, public deployment.

NEVER CUT
Policy engine, critic, seller gate, audit log, 4 demo cases.

SERVER-COMPUTED REFUND CAP

```
0 < amount <= min(₹4,999, remaining refundable amount, order paid amount)
```

Idempotent single transaction per order.

PS-04 checklist: mock tool calls, refund-limit guardrails, action log, human confirmation before refund.
No idea can guarantee top-100 selection. This deck claims a checkable, buildable agent, not a guaranteed result.